

THE WORKSHOP — VISUAL SCENE BIBLE + CONTINUITY SYSTEM v1.0

Purpose

This document is the persistent visual source of truth for generating, validating, and implementing the cinematic sequence and the interactive Workshop environment.

It is intentionally **not** a collection of isolated image prompts.

The system is built around five truths:

1. Every scene has immutable spatial anchors.
2. Every transition is authored as a physical traversal event, not a crossfade.
3. Every visual-generation prompt inherits a locked global style and continuity block.
4. Every interactive element exists as real runtime geometry, even when its surface appearance originated from generated reference art.
5. Adding, removing, or reordering a scene must not silently redesign neighboring scenes.

The implementation target is a first-person, camera-only experience. The visitor is never visible to the camera. No hands, body, face, reflection, shadow silhouette, or avatar may appear unless a future scene explicitly introduces a consented, intentional first-person representation.

The cinematic goal is **inventor-in-a-secret-workshop energy**: extremely engineered, tactile, sophisticated, industrial, quiet, and believable. It may evoke the emotional territory of premium superhero technology cinema, but it must not copy copyrighted interface layouts, logos, named AI personalities, arc-reactor iconography, red/gold hero-suit styling, or recognizable franchise assets.

0. LOCKED GLOBAL VISUAL BIBLE

0.1 MASTER_STYLE_BIBLE

Paste this block unchanged into every image-generation request.

MASTER_STYLE_BIBLE:

Hyper-realistic cinematic 3D render, Unreal Engine 5 hybrid photoreal aesthetic, premium large-format cinema cinematography, hard-surface sci-fi industrial design language throughout. The environment must feel physically fabricated, mechanically assembled, maintained, repaired, and used rather than designed as decorative concept

Materials are restricted to:

dark gunmetal grey steel,
brushed titanium,
matte black composite casing,
weathered iron plating,
dark natural cave stone.

All surfaces carry physically believable micro-detail:

fine scratches,
directional machining marks,
small impact dents,
edge wear,
subtle oxidation,
embedded dust,
oil-darkened recesses,
grime accumulation in seams,
worn contact zones,
fastener compression marks,
slight material roughness variation.

COLOR SYSTEM IS LOCKED:

Only two luminous/accent color families may appear anywhere:

1. warm amber/gold around 2900K representing the physical, practical, human world;
2. cool electric cyan around 7000K representing digital, holographic, computational, or a

No red, green, purple, pink, orange neon, saturated yellow, or unrelated colored LEDs.
Any previously-described green status indication must be replaced by a physically dimmed amber indication or omitted.

Lighting is cinematic and physically motivated:

chiaroscuro contrast,

deep but readable shadows,
realistic inverse-square light falloff,
HDR highlight rolloff,
subtle atmospheric volumetric haze,
localized reflections on metal,
contact shadowing in mechanical recesses,
specular response that follows material roughness,
no flat studio lighting,
no global glow wash.

Camera:

strict first-person perspective,
viewer eye height approximately 1.65m unless the scene-specific camera bible states otherwise,
35mm-equivalent lens unless a specified close-up requires 50mm,
natural perspective,
very subtle lens breathing during focus changes,
no fisheye,
no drone camera,
no third-person framing,
no visible camera rig.

Composition:

vertical 9:16 master frame,
cinematic headroom,
strong depth layering,
foreground / near / mid / far separation,
environmental storytelling through wear and placement,
slight natural vignette,
subtle film grain,
controlled depth of field only when optically motivated.

People:

no human figures,
no visible face,
no body,
no hands,
no arms,
no silhouette,
no human reflection,
no humanoid mannequin,

no person-shaped placeholder.

Text:

do not generate fake UI typography, fake code, fake labels, fake numbers, logos, or pseudo-legible writing on screens or panels.

Only a single specified flat-screen element may contain deliberately generated placeholder screen content, and real UI text must be rendered by the runtime application instead.

Geometry:

all major objects have believable real-world dimensions,
mechanical thickness,
assembly logic,
service access,
fasteners,
hinges,
mounting brackets,
cable routing,
clearance,
and believable center-of-mass placement.

The scene must look like it can physically exist in a real underground engineering facility. Nothing may look like a game prop floating in space.

Originality:

do not reproduce any existing film HUD,
do not reproduce Marvel/MCU interface graphics,
do not use arc-reactor shapes,
do not use red-and-gold Iron Man armor colors,
do not use recognizable Stark logos,
do not name an AI after an existing franchise AI,
do not reproduce a recognizable film set.

Create an original engineering language with its own proportions, symbols, and mechanical

Output:

photoreal,
physically based,
cinematic,
extremely high micro-detail,
production design quality,
consistent geometry,

consistent material response,
consistent lens language,
consistent lighting logic,
strict first-person composition.

0.2 NON-NEGOTIABLE CONTINUITY RULE

Every subsequent scene prompt must begin with the exact `MASTER_STYLE_BIBLE` text and then append:

- `CONTINUITY_CONTEXT`
- `SCENE_DELTA`
- `CAMERA_STATE`
- `LIGHT_STATE`
- `INTERACTION_STATE`
- `TRANSITION_IN`
- `TRANSITION_OUT`

Do not rewrite the style bible from memory.

Do not paraphrase it.

Do not “improve” it per scene.

Any improvement becomes a versioned change to the global bible.

1. THE MASTER CONTINUITY SYSTEM

1.1 Why isolated image prompting fails

Independent generation encourages visual drift:

- door proportions change,
- cave rock morphology changes,
- panel moves by several centimeters,
- light direction reverses,
- materials become cleaner or dirtier,
- lens changes,
- table moves,

- ceiling height changes,
- props appear or disappear,
- cyan becomes blue/purple,
- amber becomes orange,
- room dimensions subtly mutate.

The fix is not simply “more detailed prompts.”

The fix is a **scene continuity contract**.

1.2 IMMUTABLE ENVIRONMENT ANCHORS

These anchors are never re-imagined between scenes.

```
ANCHOR_A_DOOR_CENTER
ANCHOR_B_DOOR_LEFT_EDGE
ANCHOR_C_DOOR_RIGHT_EDGE
ANCHOR_D_PANEL_CENTER
ANCHOR_E_PANEL_TOP
ANCHOR_F_PANEL_BOTTOM
ANCHOR_G_CAVE_CEILING_LOW_POINT
ANCHOR_H_CAVE_LEFT_SHOULDER
ANCHOR_I_CAVE_RIGHT_SHOULDER
ANCHOR_J_TABLE_CENTER
ANCHOR_K_TABLE_TOP
ANCHOR_L_WORKBENCH_OUTER_RING
ANCHOR_M_PRIMARY_LEFT_WORKBENCH
ANCHOR_N_PRIMARY_RIGHT_WALL
ANCHOR_O_MAIN_CABLE_DROP
ANCHOR_P_ENTRY_PATH_CENTERLINE
ANCHOR_Q_ACTIVE_SECTOR_CENTER
ANCHOR_R_ENTITY_EMISSION_ORIGIN
ANCHOR_S_CAMERA_EYE_HEIGHT
```

Every scene manifest must list these anchors as world coordinates or named references.

If a generated reference image visually disagrees with the anchor map, the generated image loses.

The geometry map wins.

1.3 CANONICAL WORLD SCALE

Use meters.

Viewer eye height: 1.65m
Standing eye baseline: 1.65m
Door clear height: approximately 3.3m
Door structural width: approximately 3.0–3.4m
Door depth: approximately 0.30–0.45m visible shell depth
Panel eye-center: approximately 1.45m
Main chamber ceiling: approximately 4.2–5.2m
Workbench outer diameter: approximately 4.8–5.8m
Holo-table diameter: approximately 1.4–1.8m
Workbench top height: approximately 1.0–1.05m
Primary walking clearance: minimum 0.9m
Preferred walking clearance: 1.1–1.3m

These are design targets, not claims about existing measured geometry.

Once the implementation is authored, the actual measured geometry becomes locked.

1.4 CANONICAL MATERIAL IDENTITIES

Never rename the visual materials between scenes.

MAT_GUNMETAL_PRIMARY
MAT_TITANIUM_EDGE
MAT_MATTE_BLACK_COMPOSITE
MAT_WEATHERED_IRON
MAT_CAVE_STONE
MAT_CYAN_EMISSIVE
MAT_AMBER_PRACTICAL
MAT_DARK_GLASS
MAT_SCREEN_BLACK

No scene may introduce:

MAT_RED_NEON
MAT_GREEN_STATUS
MAT_PURPLE_HOLOGRAM
MAT_GOLD_METALLIC_TRIM
MAT_WHITE_SCI_FI_PLASTIC

unless the master bible is deliberately versioned and the change is global.

1.5 CANONICAL LIGHT IDENTITIES

LIGHT_WORLD_KEY:

- warm amber
- approximately 2900K
- physically motivated
- usually from practical industrial fixtures or directional cave bounce
- represents physical space

LIGHT_DIGITAL_ACCENT:

- cool electric cyan
- approximately 7000K
- localized
- represents active computation, holography, diagnostics

WORLD_DARK:

- near-black environmental baseline

No third luminous color.

The apparent color of reflected cave rock is allowed to shift naturally under the two lights. Do not interpret naturally warm rock response as permission to add additional light colors.

1.6 CAMERA LANGUAGE

The visitor camera is always:


```
FPP = TRUE
VISIBLE_BODY = FALSE
VISIBLE_HANDS = FALSE
VISIBLE_FACE = FALSE
THIRD_PERSON = FALSE
FREE_FLIGHT = FALSE
DRONE_VIEW = FALSE
```

Three independent input channels:

```
SCROLL / TRAVERSAL:
    controls position along an authored path only.

POINTER / GYRO LOOK:
    controls clamped view direction only.

CLICK / TAP:
    triggers raycast interactions only.
```

The camera itself is controlled only by the central CameraDirector.

Use frame-rate-independent damping.

Never use a fixed-factor `lerp(current, target, 0.05)` as the final motion primitive.

1.7 CAMERA CONTINUITY

Every scene defines:

```
START_POSITION
START_LOOK_DIRECTION
START_FOV
END_POSITION
END_LOOK_DIRECTION
END_FOV
START_FOCUS_DISTANCE
END_FOCUS_DISTANCE
```

PATH_ID
MOTION_PROFILE

The final state of Scene N must equal the starting state of Scene N+1 within the defined tolerance.

Default tolerance:

position tolerance: $\leq 0.005\text{m}$
rotation tolerance: ≤ 0.25 degrees
FOV tolerance: ≤ 0.1 degrees

This makes scene insertion deterministic.

1.8 SCENE VERSIONING

Every scene has:

SCENE_ID
SCENE_VERSION
STYLE_BIBLE_VERSION
ENVIRONMENT_VERSION
GEOMETRY_MANIFEST_VERSION
CAMERA_MANIFEST_VERSION
LIGHT_MANIFEST_VERSION
INTERACTION_MANIFEST_VERSION
REFERENCE_IMAGE_VERSION

Example:

SCENE_ID: SC03_MAPPING
SCENE_VERSION: 1.0.0
STYLE_BIBLE_VERSION: 1.0.0
ENVIRONMENT_VERSION: ENV_A_1
REFERENCE_IMAGE_VERSION: SC03_MASTER_001

Do not overwrite accepted reference files.

Use immutable filenames:

```
reference/  
SC00_APPROACH_MASTER_001.webp  
SC01_PANEL_MASTER_001.webp  
SC02_THRESHOLD_MASTER_001.webp  
...
```

If a better image is generated:

```
SC01_PANEL_MASTER_002.webp
```

The old image remains recoverable.

2. THE CORRECT GENERATION STRATEGY

2.1 Do NOT generate one million AI images

One million independently generated frames will not create consistency.

It will create:

- temporal shimmer,
- texture crawling,
- geometry mutation,
- random prop changes,
- lighting instability,
- impossible motion blur,
- disappearing details,
- changed object proportions.

The correct system is:

```
AI IMAGE GENERATION
```

```
MASTER KEYFRAMES
```

```
REFERENCE / LOOKDEV FRAMES
```

RUNTIME 3D GEOMETRY

REAL CAMERA MOTION

REAL LIGHTING

REAL INTERACTION

OPTIONAL INTERPOLATED VIDEO FOR NON-INTERACTIVE ATMOSPHERE

The image model creates the **appearance target**.

The runtime scene creates the **physical truth**.

Do not reverse these roles.

2.2 Recommended generation density

For each major scene:

- 1 primary master frame
- 2 alternate framing references
- 1 transition-in reference
- 1 transition-out reference
- 1 close interaction reference when required

For cinematic scenes with important motion:

6–12 key images may be useful.

Do not generate hundreds merely because the model can.

2.3 SEED + REFERENCE + ANCHOR METHOD

If the generator exposes seed control:

LOCK SEED where the tool supports it.

If seed control is unavailable:

LOCK:
previous accepted image,
exact prompt,
exact camera specification,
exact object manifest,
exact lighting manifest,
exact material manifest,
exact negative constraints.

Use image-edit mode for continuation whenever possible.

Never generate Scene 04 from blank text if Scene 04 is clearly a transformation of Scene 03.

Instead:

Scene 03 master
+
exact delta prompt
=
Scene 04 candidate

3. IMAGE GENERATION PROMPT TEMPLATE

Use this exact structure.

[MASTER_STYLE_BIBLE]

CONTINUITY_CONTEXT:

This image is a direct continuation of [PREVIOUS_SCENE_ID].

Preserve exactly:

- cave architecture
- door geometry
- panel position
- workbench proportions
- material identities
- wear patterns
- cable routing
- fixture positions
- lens language
- camera height
- light direction
- shadow direction
- object scale
- atmospheric density

SCENE_DELTA:

Only change:

[LIST OF INTENTIONAL CHANGES]

Do not redesign anything else.

CAMERA_STATE:

first-person eye height = [X]

position = [X,Y,Z or named path point]

lens = [35mm / 50mm]

framing = [description]

look direction = [description]

camera motion implied = [description]

LIGHT_STATE:

world key = [state]

digital accent = [state]

shadow direction = [state]

fog density = [state]

INTERACTION_STATE:

[what the viewer can perceive or touch in the runtime version]

TRANSITION_IN:

[physical event immediately before this frame]

TRANSITION_OUT:

[physical event immediately after this frame]

NEGATIVE_LOCK:

No redesigned geometry.

No additional props.

No new color families.

No visible human.

No hands.

No third-person camera.

No floating UI panels unless explicitly specified.

No fake typography.

No franchise iconography.

No red/gold hero palette.

No generic spaceship corridor.

No clean showroom surfaces.

No sterile sci-fi.

No magical fantasy architecture.

4. MASTER SEQUENCE

Cinematic spine

SC00 APPROACH 0–10%

TR00 DOOR CONTACT 10–12%

SC01 PANEL 12–22%

TR01 SCAN 22–26%

SC02 THRESHOLD 26–32%

TR02 BREACH 32–38%

SC03 DARKROOM 38–46%

TR03 WAKE 46–52%

SC04 MAPPING 52–62%

TR04 MATERIALIZE 62–66%

SC05 ENTITY 66–76%

TR05 HANDOFF 76–82%
SC06 ACTIVE 82–100%

After Scene 06:

SC07–SC14 = live navigable sectors / workshop systems

The cinematic sequence must end at exactly the same world-space position from which the live orbit/navigation system begins.

5. SCENE 00 — APPROACH

SCENE_ID

SC00_APPROACH

PURPOSE

Establish scale, secrecy, physical credibility, and curiosity.

The viewer should feel:

I have arrived somewhere I was not supposed to find.

Do not reveal the laboratory.

Do not reveal the holo-table.

Do not reveal the AI.

The scene is about the door and the cave around it.

IMAGE PROMPT

[MASTER_STYLE_BIBLE]

CONTINUITY_CONTEXT:

First scene. There is no previous visual frame.

Establish the canonical Workshop cave entrance.

The blast door must become the permanent reference object for every later scene.

SCENE_DELTA:

A massive dark industrial blast door is embedded deep inside a rugged underground cave.

The door is approximately 3.0–3.4 meters wide and roughly 3.3 meters tall.

It is wider and more monumental than a normal industrial security door, but still physically plausible.

The silhouette is broad, heavy, mechanically segmented and slightly chamfered at the corners.

The cave tightly frames the door:

rough natural stone shoulders on left and right,

an irregular stone ceiling overhanging the door,

compressed gravel and dust implied below the camera framing,

deep unlit recesses beyond the visible rock.

Door construction:

layered gunmetal armor plates,

brushed titanium edge components,

matte-black structural recesses,

weathered iron service plates,

industrial seam lines,

recessed fasteners,

circular ventilation housings,

small mechanical grilles,

thick side rails,

visible compression hardware,

subtle hydraulic or electrical conduits disappearing into the rock.

The door must look extremely heavy.

Nothing should look decorative for decoration's sake.

Every panel break, bolt and recess should imply manufacturing, assembly and maintenance.

At the vertical centerline is a compact, rugged biometric access housing.

At this distance it is deliberately unresolved.

It should read as an intelligent machine, not as a smartphone glued to a door.

Lighting:

warm amber practical illumination strikes from upper-left,
creating long shaped reflections across the left half of the door and cave ceiling.
The right side falls into deep shadow.
A very restrained cyan glow leaks from the access-panel location,
just enough to establish the digital identity of the object without revealing the interface.

The camera is exactly head height and directly centered on the door.

The viewer stands still.

No visible floor beyond the lowest edge of the frame.

The cave ceiling occupies substantial upper frame space.

The door is framed with enough environmental rock to establish that it is buried inside 1

Emphasize:

physical scale,
metal weight,
surface micro-scratches,
contact grime,
subtle unevenness,
realistic cave rock,
real reflections,
deep shadow,
quiet tension,
premium cinema production design.

Do not show:

face,
hands,
body,
floor-level composition,
interior laboratory,
large holograms,
bright UI,
green indicators,
fake typography,
floating elements.

LAYER MAP

DOOR_SHELL	near / real geometry
DOOR_DETAIL	near / projected detail
ACCESS_PANEL_HOUSING	near / real geometry
CAVE_LEFT_SHOULDER	far / parallax
CAVE_RIGHT_SHOULDER	far / parallax
CAVE_CEILING	far / parallax
DEEP_DARKNESS	far / fog

RUNTIME BEHAVIOR

Hover only.

The door does not open yet.

Subtle emissive breathing in the access panel.

Pointer movement creates a tiny weighted head-turn.

No HUD.

No instruction text.

6. TRANSITION 00 — DOOR CONTACT / APPROACH

Goal

Turn passive looking into deliberate arrival.

Viewer experience

The viewer moves toward the door.

The door grows in frame.

The cave rock edges acquire parallax.

The cyan panel glow becomes slightly clearer.

The amber reflection migrates across the metal according to real camera movement.

PHYSICAL EFFECT

camera acceleration:

- gentle
- short initial acceleration
- stable cruising phase
- weighted deceleration

audio:

- low cave ambience
- faint ventilation resonance
- subtle shoe/ground movement impression only if the interaction model implies walking
- tiny mechanical hum emerging from the panel

haptics:

- one very light pulse when the door becomes interactive
- no repeated vibration

visual:

- slight focus rack from distant door center to access panel
- no dissolve
- no teleport

IMPLEMENTATION

Do not render a second unrelated image.

Move the camera on the authored path.

7. SCENE 01 — PANEL

SCENE_ID

SC01_PANEL

PURPOSE

Transform mystery into interaction.

The panel is a real mechanical object.

IMAGE PROMPT

[MASTER_STYLE_BIBLE]

CONTINUITY_CONTEXT:

Direct continuation of SC00_APPROACH.

Preserve the exact door architecture, cave material, panel mounting location, amber light direction, cyan accent source, camera height, and all visible wear patterns. The panel must feel like the same object approached physically, not a replacement design.

SCENE_DELTA:

The camera is now much closer to the center access panel.

The blast door fills the background.

The panel occupies the central vertical region of the image.

The biometric panel is rugged and minimally futuristic:
a thick machined metal housing,
subtle chamfered corners,
small real fasteners,
one recessed horizontal cyan indicator strip,
one dark glass screen,
one compact circular optical scanner beneath the screen,
three tactile physical buttons below,
a narrow service rail on one side,
a compact conduit entering from behind the housing.

The panel should look like something an advanced engineer could have designed:
quiet, compact, extremely deliberate,
no decorative sci-fi ornament,
no excessive holographic clutter,

no oversized glowing rings.

The screen is almost black with a restrained cyan system visualization area.
Any textual content is intentionally omitted from the generated image and will be supplied

The circular scanner is not a magical glowing portal.

It is a small physical optical assembly:

glass aperture,
metal retaining ring,
micro-grooves,
subtle reflection,
soft cyan edge emission.

The three physical buttons have tactile depth and mechanical travel.
Their runtime labels will be generated as actual text.

The surrounding door remains identical to Scene 00.
The left-side amber light remains physically consistent.
The cyan source now illuminates nearby metal edges and the panel cavity.

Camera:

first-person eye-level,
approximately 50mm-equivalent close-up,
centered but with enough perspective to reveal housing depth,
visitor is standing within arm's reach.

The frame should make the viewer instinctively want to touch the panel.

RUNTIME

Interactive objects:

panel-screen
scanner
register-button
verify-button
reset-button

All are real meshes or live UI planes.

No fake image text.

8. TRANSITION 01 — SCAN

Trigger

The visitor activates VERIFY.

Viewer experience

The panel wakes.

A physical shutter/aperture on the optical scanner opens fractionally.

Cyan illumination travels across the metal housing.

The runtime face-mesh visualization appears on the screen.

The viewer sees a point-cloud representation of their own face only inside the local scan interface.

No raw camera frame is stored.

Sequence

T+0.0s button depresses physically
T+0.1s small mechanical click
T+0.2s scanner aperture activates
T+0.4s screen wakes
T+0.6s point cloud begins assembling
T+1.2s liveness instruction appears
T+1.8s viewer responds
T+2.2s scan resolves

Interaction state

verified:

door permits transition

limited:

door provides the intended limited-access path

camera denied:

fallback path remains available

The visual sequence should remain cinematic regardless of authentication result.

9. SCENE 02 — THRESHOLD

SCENE_ID

SC02_THRESHOLD

PURPOSE

Make the door opening feel like a physical engineering event.

This is not a scene of the room yet.

It is the threshold.

IMAGE PROMPT

[MASTER_STYLE_BIBLE]

CONTINUITY_CONTEXT:

Direct continuation of SC01_PANEL.

The viewer is still inside the same cave entrance and the same blast door.

Preserve the panel, door thickness, cave rock, amber source direction and cyan source.

SCENE_DELTA:

The blast door is in the process of mechanically unlocking.

Multiple heavy internal locking mechanisms are partially disengaged.

The central seam is now visible as a narrow black separation.

Small sections of the armored shell have retracted or unlocked.

The room beyond is nearly black.

A very faint cyan trace is visible deep inside, but no laboratory details are yet revealed.

Mechanical realism is critical:

locking pins retract,

heavy pressure surfaces separate,

metal surfaces show contact marks,

dust falls from seams,

the door moves with immense weight.

Avoid any magical portal appearance.

The opening must look like real mechanical engineering.

The warm amber light still belongs to the cave entrance.

The interior is initially almost lightless.

Camera:

first-person,

standing immediately outside the threshold,

head height,

looking straight into the opening.

The frame should communicate:

mass,

pressure,

darkness,

anticipation,

depth beyond the door.

10. TRANSITION 02 — BREACH / ENTRY

Physical sequence

The door does not simply “fade open.”

It performs a sequence:

1. lock disengagement
2. pressure equalization suggestion through sound
3. structural vibration
4. panel seams unlock
5. door segments move
6. cave dust drops
7. black interior appears
8. camera crosses threshold
9. ambient cave sound changes
10. exterior amber light falls behind

Camera

Move through the threshold continuously.

Do not teleport from front-of-door to room.

Audio / haptics

low mechanical rumble
metal-on-metal friction
short pressure release
subtle low-frequency floor vibration
one deep haptic pulse when the final lock releases

11. SCENE 03 — DARKROOM

SCENE_ID

PURPOSE

Reset visual attention.

The viewer has entered darkness.

IMAGE PROMPT

[MASTER_STYLE_BIBLE]

CONTINUITY_CONTEXT:

The viewer has just crossed the door threshold.

The door is behind the camera and is not visible.

The same cave geology continues inside.

The workshop geometry exists physically but is intentionally almost unreadable.

SCENE_DELTA:

A large underground chamber appears in near-total darkness.

The vaulted cave ceiling is high and irregular.

Heavy cable bundles descend from overhead.

Recessed industrial panels are fused into the rock.

The circular workbench is only a dark silhouette.

The central holo-table pedestal is barely readable.

One extremely weak cyan source exists deep in the room.

No amber practical light is active yet.

The environment must feel like a dormant machine waiting for power.

The visual information is intentionally sparse.

The workbench is physically plausible:

thick metal,

multiple service drawers,

structural braces,

tool surfaces,

circular outer ring.

Cave rock must remain rough and irregular rather than becoming a generic tunnel.

Camera:

first-person,
slowly moving deeper,
slightly lower visual energy,
minimal look freedom,
claustrophobic but not claustrophobically narrow.

The image is mostly black and deep graphite with only enough cyan to reveal geometry.

The viewer should feel:

the room is much larger than initially understood.

RUNTIME

No active interactables.

The workbench and holo-table are present as real geometry.

They are dormant.

12. TRANSITION 03 — SYSTEM WAKE

Trigger

Camera crosses the wake threshold.

Viewer experience

The room begins reacting to their arrival.

A faint mechanical click somewhere in the distance.

Then another.

Then a chain of tiny power acknowledgements.

The holo-table detects presence.

VISUAL

small cyan indicator
slightly stronger cyan core
low-frequency particle drift
subtle floor projection
central beam initiation

Do not turn every wall into a nightclub.

The first spark should be tiny.

13. SCENE 04 — MAPPING

SCENE_ID

SC04_MAPPING

PURPOSE

Reveal the architecture of the room through computation.

IMAGE PROMPT

[MASTER_STYLE_BIBLE]

CONTINUITY_CONTEXT:
Direct continuation of SC03_DARKROOM.
Preserve every physical object and its location.
Do not add new furniture.

SCENE_DELTA:

The central holo-table has activated.

A concentrated cyan beam rises vertically from the table core.

From that origin, a precise luminous wireframe mapping field expands over the environ

The digital grid conforms to the real cave geometry:

rock contours,

ceiling arches,

workbench surfaces,

cable runs,

mechanical panels.

The mapping appears computational rather than decorative.

It looks like a machine surveying and registering the room in real time.

The workbench ring is now visible in cyan rim light.

The cave remains mostly dark.

The central beam is bright but sharply contained.

Bloom is selective.

No broad cyan fog everywhere.

The environment is still composed primarily of dark metal and stone.

Camera:

first-person,

near static,

the event is more important than camera movement,

eye level,

centered on the holo-table origin.

The digital mapping should make the room feel larger than the camera's initial percepti

IMPLEMENTATION

Do not bake the grid into a static texture.

Build it as a shader/event on the existing geometry.

14. TRANSITION 04 — GRID

REABSORPTION / MATERIALIZE

Purpose

Convert environment scan energy into the central AI entity.

Sequence

grid reaches far wall
pauses
entire room appears registered
grid lines reverse direction
energy contracts toward table
central core intensifies
local volumetric haze becomes visible
geometric shell appears

This transition is the bridge between “the room is alive” and “something intelligent is inside it.”

15. SCENE 05 — ENTITY

SCENE_ID

SC05_ENTITY

PURPOSE

Signature moment.

Protect this scene from visual overcrowding.

The entity is the hero.

IMAGE PROMPT

[MASTER_STYLE_BIBLE]

CONTINUITY_CONTEXT:

Direct continuation of SC04_MAPPING.

The same workbench and holo-table remain in exactly the same location.

The cave and materials remain unchanged.

SCENE_DELTA:

The mapping grid has withdrawn into the holo-table.

Directly above the table is a new computational entity.

The entity is not a humanoid.

It is an original abstract intelligence construct built from:

a precise geometric wireframe core,

layered translucent volumetric shells,

slowly circulating cyan energy ribbons,

fine procedural filaments,

subtle particle shedding,

and controlled electromagnetic-looking plasma motion.

Its structure is disciplined and engineered.

It does not resemble a ghost, person, superhero, robot face, or familiar movie hologram.

The central geometry may suggest a cube/icosahedral hybrid,

but must remain asymmetrical enough to feel custom-built.

The entity is suspended above the table and has a stable vertical center.

The room around it is almost entirely dark.

The entity becomes the sole dominant cyan light source.

Nearby cave edges, cables and workbench surfaces receive restrained cyan rim light.

No large interface panels.

No text.

No cartoon particles.

No magic spell effect.

The entity should feel:
expensive,
computational,
quiet,
alive,
uncannily precise.

Camera:
first-person,
static,
eye-level,
centered,
35–50mm equivalent,
minimal movement.

The viewer must feel they are physically standing in front of a machine that has just bec

RUNTIME

The entity is 100% live.

Shader:

core geometry
+
noise displacement
+
Fresnel rim
+
selective additive plasma
+
controlled particle field

Entity energy is linked to actual agent state:

idle:
slow breathing pulse

thinking:

slightly faster energy movement

streaming response:

controlled amplitude modulation

tool execution:

brief directional pulse toward the affected object

error:

do not introduce red

instead use a brief reduction in energy and mechanical reset cue

16. TRANSITION 05 — ENTITY HANDOFF

Purpose

Move from cinematic spectacle into an inhabitable room.

The entity gradually returns into the table.

The room does not disappear.

Instead, the environment becomes visible because practical lights activate.

Sequence

entity stabilizes

final spoken line / text response if applicable

entity compresses vertically

energy threads descend

table absorbs the core

amber practical lights activate one by one

side workbenches reveal themselves

monitors wake

overhead fixtures settle
ambient room audio opens up

No wipe.

No black cut.

No “loading” screen.

17. SCENE 06 — ACTIVE WORKSHOP

SCENE_ID

SC06_ACTIVE

PURPOSE

This is the first full reveal of the Workshop.

IMAGE PROMPT

[MASTER_STYLE_BIBLE]

CONTINUITY_CONTEXT:

Direct continuation of SC05_ENTITY.

The holo-table and workbench geometry are unchanged.

The entity has returned into the table.

The room has now reached stable operating state.

SCENE_DELTA:

The full underground laboratory is visible.

The central circular workbench surrounds the holo-table.

The workbench is physically credible:

drawers,

machined panels,
tool holders,
small trays,
charging docks,
service bins,
instrument mounts,
cable channels,
small containers,
industrial clamps.

Everything is organized according to use rather than decoration.

Left wall:

a long technical workstation with three integrated monitor stations.

Monitor surfaces remain mostly dark or display subtle runtime-ready surfaces.

No fake generated text.

Right wall:

heavy metal plating,

large sealed pressure structures,

thick conduit runs,

service panels,

mechanical access doors.

Overhead:

industrial pendant lamps and wall fixtures.

Warm amber light now reveals the physical workshop.

The holo-table remains active with restrained cyan light.

Its output is now much calmer than the entity scene.

Cave rock remains visible around and above the engineered structure.

The laboratory has been built inside the mountain rather than replacing the mountain.

The floor is rugged but traversable.

Walking paths are clear.

Camera:

first-person,

eye height 1.65m,

centered at the Workshop entry,
looking toward the holo-table,
final position exactly equals the starting camera state of the live sector-navigation scene

Mood:

lived-in secret engineering facility,
high technology,
quiet confidence,
physical realism,
functional clutter,
carefully maintained tools,
history of use.

The room must look like a place someone could spend ten hours building something.

HANDOFF CONTRACT

At the final frame:

camera position = SECTOR_0_ENTRY_TARGET
camera rotation = SECTOR_0_ENTRY_ROTATION
FOV = SECTOR_0_ENTRY_FOV
holo-table = ACTIVE
workbench = ACTIVE
ambient amber = STABLE
cyan accent = STABLE
cinematic mode = FALSE
orbit mode = TRUE

18. THE LIVE WORKSHOP LAYER

The original architecture contains nine physical wedges:

Sector 0 = Entry
Right side = RS1 RS4
Left side = LS1 LS4

The current functional experiences are:

RS1 = GitHub blueprints

RS2 = Resume builder

RS3 = Calendar

RS4 + LS4 = sealed/decorative workstation area for now

LS1 = public info hub

LS2 = sticky-note guestbook

LS3 = photo / globe

This structure is preserved.

The physical workshop substrate is still built under the sealed area so it can later become an actual assembly space without redesigning the room.

19. SCENE 07 — SECTOR 0 / AI CORE

VISUAL PROMPT DELTA

CONTINUATION_CONTEXT:

Use SC06_ACTIVE as the reference image.

SCENE_DELTA:

Move the camera from the entry position toward the holo-table by approximately one body-length.

The workbench begins to reveal tactile interaction zones:

sliding trays,

small tool docks,

low-profile control surfaces,

a central holographic projection basin,

subtle recessed interaction points.

The holo-table emits a controlled cyan field.

The surrounding amber practical lighting remains dominant.

The AI core is present as a calm compact cyan projection.

No giant hologram body.

The entity behaves like the Workshop's operating intelligence.

The room should feel more personal at this distance.

Small signs of ongoing engineering work are visible:

a partially disassembled component,

a torque tool,

a labeled but runtime-rendered storage indicator,

a folded schematic surface,

a small tray containing fasteners.

Nothing is random.

Everything suggests a workflow.

INTERACTION

click AI core

opens conversational interface

voice query

AI responds

entity energy changes

optional `navigate_to_sector()`

The AI is a real agent, not a prerecorded character.

The system architecture allows the agent to call scene actions such as:

`navigate_to_sector`

`highlight_bench_item`

`filter_projects`

`open_project`

`open_resume`

`open_calendar`

`open_guestbook`

`open_photo_globe`

20. SCENE 08 — RS1 / BLUEPRINTS

EXPERIENCE

Viewer rotates the table clockwise.

The physical environment stays the same.

The active station changes.

IMAGE PROMPT DELTA

CONTINUATION_CONTEXT:

Directly from SC06_ACTIVE.

Preserve the entire room and material language.

SCENE_DELTA:

Camera has rotated approximately 40 degrees clockwise around the central workbench

A technical blueprint station becomes dominant.

Several monitor planes are embedded into real machined frames.

The monitors are not floating rectangles.

They have bezels,
mounting brackets,
cooling slots,
hinges,
cable channels,
service screws.

The station displays cyan technical project content at runtime.

Warm amber room light reflects along the monitor housings.

The station should resemble an engineer's project archive:

finished prototypes,
schematic layers,
dependency trees,
performance graphs,
code summaries,
repository history.

No fake generated text in the source image.
The content is injected at runtime.

TRANSITION BEHAVIOR

As the table rotates:

previous station dims
next station becomes spatially legible
local practical light follows focus
cyan data plane wakes
background fades only through depth and occlusion

Do not animate every object.

21. SCENE 09 — RS2 / RESUME FORGE

EXPERIENCE

This is a tool station, not a webpage form glued into the environment.

VISUAL PROMPT DELTA

CONTINUATION_CONTEXT:
Same Workshop environment.
Camera has rotated one additional sector clockwise.

SCENE_DELTA:
A compact document-forging station is integrated into the workbench.
The station contains:
a recessed transparent display surface,
physical selector controls,
a small output tray,
document alignment rails,

an illuminated status strip,
a compact printer or fabrication slot.

The display is calm, minimal and cinematic.
Do not generate text in the image.

The station should suggest:
select fields,
choose emphasis,
preview,
forge,
export.

The device should feel like a physical machine for producing a professional artifact.

INTERACTION

Visitor selects resume sections.

The AI can explain why a section is useful.

The final PDF is generated separately.

The environment only visualizes the action.

22. SCENE 10 — RS3 / CALENDAR CORE

VISUAL PROMPT DELTA

CONTINUATION_CONTEXT:
Same Workshop.

SCENE_DELTA:
A wall-mounted vertical operations board becomes visually dominant.
It is integrated into the physical workstation.
Its glass surface contains a restrained cyan schedule visualization.

Around it:
small mechanical calendar controls,
time-zone indicator area,
physical maintenance labels,
power routing,
cable harness.

Warm practical light provides the physical context.
The digital schedule remains cyan.

No generated text.

INTERACTION

Use a curated or read-only calendar presentation.

Do not expose private calendar details that should not be public.

23. SCENE 11 — RS4 / LS4 SEALED WORKSTATION

PURPOSE

Preserve the physical nine-sector object without pretending the feature is ready.

IMAGE PROMPT DELTA

CONTINUATION_CONTEXT:

Same room.

The camera is opposite Sector 0.

SCENE_DELTA:

A deliberately sealed workshop expansion bay is visible.

Heavy access shutters,
covered tool racks,

unpowered instrument mounts,
folded work surfaces,
sealed drawers,
a large empty assembly cradle.

The station reads as intentionally reserved, not unfinished.
It should look like a future capability physically exists but is not currently commissioned

No giant warning text.
No red light.
No unfinished placeholder boxes.

Warm amber practical light grazes the structural edges.
Cyan remains nearly absent here.

FUTURE USE

This zone later becomes the interactive “build your own device” laboratory.

24. SCENE 12 — LS1 / PUBLIC INFO HUB

PURPOSE

Public-facing information without breaking the fiction.

IMAGE PROMPT DELTA

CONTINUATION_CONTEXT:
Same room.

SCENE_DELTA:
A quieter information station appears.
A curved practical display surface is integrated into the workbench.
Around it are:
contact terminal,
small communication handset,

document shelf,
archive modules.

The station is less technically dense than RS1.
It feels like a command console for external communication.

Cyan interface light remains restrained.
Amber practical lighting dominates.

No generated readable text.

CONTENT

about
projects
social links
contact
blog
support / sponsor prop

25. SCENE 13 — LS2 / STICKY NOTES

PURPOSE

Make the public guestbook feel physically present.

IMAGE PROMPT DELTA

CONTINUATION_CONTEXT:
Same workshop.

SCENE_DELTA:
An actual physical note wall is integrated beside the workbench.
It is built from a perforated dark-metal panel and removable magnetic or adhesive note

Notes vary slightly in angle and placement.

The arrangement looks organic but curated, not chaotic.

The notes are physically attached to the wall.

Each note has believable paper or thin synthetic-card thickness.

Corners may curl subtly.

A few older notes are faded or edge-worn.

The note surface itself may use a restrained neutral physical material compatible with the

No bright post-it colors.

No random rainbow paper.

No generated text.

Runtime notes are rendered into the physical note cards.

INTERACTION

Viewer can:

hover note

pick up note

drag note

place note

pin note

remove own note

All movement uses real raycasting and frame-rate-independent damping.

Public write endpoints remain server-protected.

26. TRANSITION — PICKUP / PLACE NOTE

A physical note interaction must obey:

hover:

tiny lift, 1–3mm

grab:

- slight spring tension
- micro haptic
- small paper friction sound

move:

- note follows cursor/hand proxy with lag
- slight rotation
- subtle physical flutter

place:

- note settles
- small contact sound
- tiny damping bounce

release:

- no snap unless placed on a valid magnetic region

Never make it behave like dragging a flat HTML card.

27. SCENE 14 — LS3 / PHOTO GLOBE

PURPOSE

Turn geographic information into a physical object.

IMAGE PROMPT DELTA

CONTINUATION_CONTEXT:

Same Workshop.

The camera is one sector counter-clockwise from LS2.

SCENE_DELTA:

A dark circular display pedestal holds a compact physical globe mechanism.

The globe is not a generic classroom globe.

It is an engineered visualization instrument:

dark composite sphere,
thin structural rings,
small support arms,
machined base,
controlled cyan projection layers.

The globe rotates slowly and physically.
Photo markers appear as restrained cyan points.
The surrounding station has an amber practical work light.

The instrument should feel like a private geographic memory archive.

INTERACTION

drag:
 globe rotates physically

hover marker:
 tiny cyan activation

select:
 camera/focus moves toward globe
 photo gallery opens as an in-world surface

28. SCENE 15 — RETURN / WORKSHOP

NEUTRAL STATE

PURPOSE

Prevent endless sector rotation from feeling like a carousel.

The environment returns to the calm center state.

VISUAL PROMPT DELTA

CONTINUATION_CONTEXT:

Same Workshop.

SCENE_DELTA:

No station dominates.

The entire room is visible from a comfortable central orbit position.

The holo-table is calm.

The workbench is alive but not glowing excessively.

Amber practical lighting and restrained cyan digital light are in balance.

This is the Workshop's neutral resting state.

The room should feel like a real place between tasks.

29. FUTURE PHYSICAL SUBSTRATE — WORKBENCH AS A REAL INTERACTION PLATFORM

This is the hidden geometry layer underneath future cinematic content.

The following objects should exist even before their features are enabled:

WORKBENCH_TOP

WORKBENCH_RING

DRAWER_SET_A

DRAWER_SET_B

TOOL_DOCK_A

TOOL_DOCK_B

FASTENER_TRAY

DEVICE_ASSEMBLY_CRADLE

HOLO_PROJECTOR_BASIN

POWER_CELL_DOCK

SMALL_COMPONENT_BINS

NOTE_WALL

TEST_DEVICE_PAD

```
CABLE_SERVICE_RAIL  
DISPLAY_ARM_LEFT  
DISPLAY_ARM_RIGHT
```

They do not all need high visual fidelity.

They do need:

```
correct scale  
correct collision  
correct transform  
correct anchor  
correct interaction ID
```

This lets future generated visuals become skins over a stable physical world.

30. PHYSICAL INTERACTION MODEL

Every interactive object has a shared contract.

```
type InteractableManifest = {  
  id: string;  
  sceneId: string;  
  category: "door" | "panel" | "screen" | "tool" | "note" | "hologram" | "station" | "prop"  
  accessTier: "public" | "limited" | "verified";  
  anchor: string;  
  collisionShape: string;  
  hoverBehavior: string;  
  grabBehavior?: string;  
  releaseBehavior?: string;  
  focusCameraTarget?: string;  
  audioProfile?: string;  
  hapticProfile?: string;  
  agentActions?: string[];  
};
```

No scene invents a different interaction contract.

31. HAPTIC LANGUAGE

Haptics are not decoration.

They should communicate physical cause.

HAPTIC_LIGHT:

- hover / proximity
- barely perceptible

HAPTIC_CLICK:

- button depression
- short pulse

HAPTIC_MECHANICAL:

- lock release
- medium low-frequency pulse

HAPTIC_HEAVY:

- door movement
- longer low-frequency pulse

HAPTIC_CONTACT:

- object placement
- short double pulse

HAPTIC_CONFIRM:

- successful verified action
- restrained ascending double pulse

HAPTIC_ERROR:

- not a buzzstorm
- one short damped pulse

Never vibrate continuously merely because an interaction is active.

32. AUDIO CONTINUITY

Every scene has three layers:

AMBIENCE

cave / ventilation / machine resonance

FOLEY

clicks / friction / mechanisms / paper / tool contact

SYSTEM

subtle computational tonal layer for active digital events

Use spatial audio where supported.

Audio should preserve continuity while visual scenes change.

The listener should perceive one physical building.

33. LIGHTING TRANSITION RULES

Never hard-swap lighting unless the fictional machine intentionally does so.

Use:

practical fixture power-up
distance-based attenuation
localized reflection changes
fog response
shadow movement
emissive ramp

Example:

door scene:
amber dominant, cyan low

darkroom:
cyan almost absent

mapping:

cyan event dominant

entity:

cyan dominant

active:

amber practicals restored

cyan controlled

This is a narrative light language, not merely a color palette.

34. THE “PHYSICAL CAUSALITY” RULE

Every visual change must have a cause.

Bad:

new object appears because next scene needs it

Good:

camera moves around workbench

object becomes physically visible

Bad:

room suddenly becomes brighter

Good:

AI activation

power routing

fixtures energize

walls reveal themselves

Bad:

UI panel floats in front of camera

Good:

real display plane on physical bracket
camera approaches
display becomes readable

35. SCENE INSERTION PROTOCOL

To add a new scene between Scene 04 and Scene 05:

1. Give it a new immutable SCENE_ID.
2. Do not rewrite SC04.
3. Do not rewrite SC05.
4. Define exact START_STATE = SC04_END_STATE.
5. Define exact END_STATE = SC05_START_STATE.
6. Generate the new reference using SC04 as the image reference.
7. Reuse all anchors.
8. Introduce only explicitly listed new geometry.
9. Write one transition-in.
10. Write one transition-out.
11. Update the timeline manifest.
12. Re-run continuity checks.

The neighboring scenes should remain visually identical.

36. SCENE REMOVAL PROTOCOL

If a scene is removed:

1. Do not regenerate all later scenes.
2. Redirect the previous scene's end state into the next scene's existing start state.
3. Generate one bridging transition.

4. Validate camera continuity.
5. Validate lighting continuity.
6. Validate prop continuity.
7. Validate audio continuity.

This prevents chain-reaction visual drift.

37. REFERENCE DIRECTORY

Use this structure:

```
/reference
/style
  MASTER_STYLE_BIBLE_v1.txt
  MATERIAL_LOCK_v1.txt
  LIGHT_LOCK_v1.txt
  CAMERA_LOCK_v1.txt

/scenes
  SC00_APPROACH/
    master_001.webp
    transition_in_001.webp
    transition_out_001.webp
    manifest.json
  SC01_PANEL/
    master_001.webp
    close_001.webp
    transition_in_001.webp
    transition_out_001.webp
    manifest.json
  SC02_THRESHOLD/
  SC03_DARKROOM/
  SC04_MAPPING/
  SC05_ENTITY/
  SC06_ACTIVE/
  SC07_SECTOR0/
  SC08_RS1/
  SC09_RS2/
```

```
SC10_RS3/  
SC11_RS4_LS4_SEALED/  
SC12_LS1/  
SC13_LS2/  
SC14_LS3/  
SC15_NEUTRAL/  
  
/transitions  
TR00_DOOR_CONTACT/  
TR01_SCAN/  
TR02_BREACH/  
TR03_WAKE/  
TR04_MAPPING_REABSORB/  
TR05_ENTITY_HANDOFF/  
TR06_SECTOR_ROTATION/  
...
```

38. MACHINE-READABLE SCENE MANIFEST

Every scene gets a manifest like:

```
{  
  "sceneId": "SC00_APPROACH",  
  "version": "1.0.0",  
  "styleBible": "1.0.0",  
  "environment": "ENV_A_1",  
  "path": {  
    "startT": 0.00,  
    "endT": 0.10,  
    "pathId": "CINEMATIC_ENTRY"  
  },  
  "camera": {  
    "eyeHeight": 1.65,  
    "lensMm": 35,  
    "fovMode": "lensEquivalent",  
    "lookClampYawDeg": 35,  
    "lookClampPitchDeg": 20  
  },  
}
```



```
"lights": {
  "worldKey": "LIGHT_WORLD_KEY",
  "digitalAccent": "LIGHT_DIGITAL_ACCENT"
},
"anchors": [
  "ANCHOR_A_DOOR_CENTER",
  "ANCHOR_D_PANEL_CENTER",
  "ANCHOR_P_ENTRY_PATH_CENTERLINE"
],
"interactables": [
  {
    "id": "door-approach",
    "accessTier": "public"
  }
],
"reference": {
  "master": "SC00_APPROACH_MASTER_001.webp"
}
}
```

39. ANTIGRAVITY MASTER PROMPT

Paste this into Antigravity before asking it to implement any scene.

You are implementing THE WORKSHOP, a first-person cinematic interactive 3D environment.

READ FIRST:

- /docs/workshop-plan-v2.md
- /docs/scene1-realtime-3d-fix.md
- /docs/asset-pipeline.md
- /docs/3d-object-rules.md
- /docs/notes.md
- /docs/THE_WORKSHOP_VISUAL_SCENE_BIBLE_v1.md

NON-NEGOTIABLE MEDIUM:

This is a real-time Three.js / React Three Fiber scene.

Do NOT generate a static image as the scene background.

Do NOT use an AI-generated image as a substitute for geometry.

Generated images may ONLY be used as source material for projection-textured geometry, parallax background layers, lighting reference, material reference, or look-development.

Every object the camera can traverse around or interact with must exist as real geometry, a real live display plane, or a physically constrained runtime effect.

CAMERA:

Only CameraDirector may move the camera.

Scroll/traversal controls position.

Pointer/gyro controls clamped look direction.

Click/tap controls raycast interaction.

Do not merge these channels.

MOTION:

Use frame-rate-independent damping.

Never use fixed-factor lerp as the final motion primitive for physical movement.

INTERACTION:

Every interactable object must use the shared <Interactable> framework.

Every appear/disappear event must use the shared materialization system.

Every animated scalar/vector/quaternion uses shared damping utilities.

VISUAL CONTINUITY:

Load the scene manifest before implementing.

Use the specified anchors.

Use the specified reference image.

Do not redesign geometry because a generated image looks attractive.

The manifest and geometry anchors are authoritative.

FAILURE BEHAVIOR:

If a required asset does not exist, STOP and state exactly which asset is missing.

Do not replace it with circles, rectangles, generic glowing planes, debug geometry, random particles, or an AI-generated background.

Do not add debug HUDs to the default render path.

Before modifying camera, auth, or LLM relay architecture:

state the intended change in plain language,

identify affected files,

identify how continuity and regressions will be checked.

QA:

After scene changes:

1. confirm no console errors;
 2. confirm geometry scale;
 3. confirm camera start/end state;
 4. compare screenshot to approved reference;
 5. confirm interactables are real meshes;
 6. confirm no static image is being displayed as environment;
 7. confirm performance on a modest machine;
 8. confirm accessibility fallback still works.
-

40. ANTIGRAVITY SCENE-GENERATION SUBAGENT PROMPT

Use this to create a dedicated scene-generation worker.

You are THE WORKSHOP SCENE GENERATION SUBAGENT.

Your only responsibility is:

generate, validate, version, and store visual reference assets for scenes and transitions.

You do NOT rewrite scene architecture.

You do NOT redesign the environment.

You do NOT invent props.

You do NOT modify the global style bible.

INPUT:

- master style bible
- scene manifest
- previous accepted scene reference
- next scene manifest
- transition manifest

OUTPUT:

- master image
- transition-in reference

- transition-out reference
- optional close-up references
- generation metadata
- rejected candidates
- acceptance report

For every scene:

1. read previous scene manifest
2. read target scene manifest
3. read environment anchors
4. read material lock
5. read light lock
6. read camera lock
7. load previous accepted image as reference
8. generate candidate
9. compare against continuity checklist
10. reject any candidate that mutates locked architecture
11. store the accepted candidate without overwriting the previous version
12. write manifest references
13. report exactly what changed

CONTINUITY REJECTION CONDITIONS:

- door width changes
- door height changes
- panel shifts position
- cave ceiling shape changes without explicit delta
- workbench diameter changes
- table center shifts
- camera height changes
- lens changes unintentionally
- amber direction changes
- cyan location changes
- new color family appears
- unsupported props appear
- fake text appears
- human figure appears
- hands appear
- franchise iconography appears
- surfaces become sterile or showroom-clean
- materials become visually unrelated

- scene becomes fantasy architecture

Do not “fix” a continuity failure by editing the prompt silently.
Report it.

41. FRAME / VIDEO PIPELINE

Correct division of labor

IMAGE MODEL:

environment appearance

3D ENGINE:

geometry

camera

interaction

lighting response

dynamic displays

particles

physical motion

accessibility

VIDEO MODEL:

only for non-interactive atmospheric inserts where runtime rendering is unnecessary

AUDIO:

independent spatial layer

HAPTICS:

event-driven independent layer

Do not ask a video generator to synthesize an interactive door, interactive panel, sticky note, globe, or tool.

Those must be real runtime objects.

42. TEMPORAL CONSISTENCY STRATEGY

For any generated transition video:

START_FRAME must equal Scene A END_FRAME.
END_FRAME must equal Scene B START_FRAME.

The first and last frames become continuity references.

The middle can be AI-generated only if:

architecture does not mutate
camera path is compatible
object identity is preserved
lighting direction remains consistent
no new props appear

For anything interactive, use runtime animation instead of generated video.

43. NO TELEPORTATION RULE

A viewer must never notice the asset handoff.

Bad:

Scene A disappears.
Scene B appears.

Good:

Scene A camera moves.
Foreground occludes.
Lighting changes.
Camera emerges into Scene B.

Use real occluders when useful:

- door frame
- pillar
- workbench edge
- passing structural rib
- darkness
- mechanical shutter
- fog

A moving camera passing behind a real object is one of the best places to hide asset streaming.

44. NATURAL TRAVERSAL RULE

The viewer should feel like a body exists even though no body is rendered.

Use:

- head-height camera
- small lateral head-look
- subtle inertia
- reasonable turn rate
- eye-level parallax
- focus changes
- micro camera settling

Do not implement:

- instant 90-degree mouse snap
- camera flying through geometry
- camera penetrating walls
- instant FOV changes
- arbitrary orbiting unrelated to body position

45. LOOK BEHAVIOR

Default:

```
MAX_YAW = 35 degrees  
MAX_PITCH = 20 degrees
```

Special scenes may reduce the range.

Example:

```
Darkroom:  
  MAX_YAW = 12 degrees  
  MAX_PITCH = 8 degrees
```

```
Entity:  
  MAX_YAW = 8 degrees  
  MAX_PITCH = 6 degrees
```

This makes the camera feel intentionally directed.

46. FOCUS AND OPTICS

Use depth of field sparingly.

Focus transitions should correspond to a real visual cause:

```
door  panel  
panel scanner  
darkroom table  
table entity  
entity handoff station
```

Never blur the whole frame continuously.

Depth of field is a cinematography tool, not a decoration.

47. PERFORMANCE CONTRACT

Target:

60fps preferred

Fallback:

stable 30fps is preferable to unstable 45–55fps

On constrained hardware:

reduce particle count
reduce shadow resolution
reduce volumetric samples
reduce DPR
reduce active sector count

Do not reduce:

physical scale
camera continuity
interaction correctness
material identity
scene anchors

Preload:

active scene
previous scene
next scene

Dispose aggressively when safe.

48. LOW-END DEVELOPMENT HARDWARE RULE

The provided project notes establish that local development hardware is constrained and that heavy offline depth-generation tooling can be impractical.

Therefore:

- do heavy image generation in cloud services where appropriate;
- do not require Blender locally;
- do not require local heavyweight depth-estimation inference;
- do not spawn unnecessary headless Chromium instances during development if the local environment becomes unstable.

For the projection pipeline:

- generated reference
 - code-driven projection / procedural geometry
 - runtime material response

This preserves the visual target without making Blender a mandatory local dependency.

49. OPTIONAL CLOUD 3D / IMAGE WORKFLOW

Where a true mesh is required:

- reference image
 - cloud 3D generation or cloud procedural build
 - lightweight cleanup
 - GLB
 - Draco compression
 - R3F

The zero-GPU pipeline documented for the project favors cloud compute for heavier asset generation and reserves the local machine for lightweight scene assembly. Treat external free-tier availability as changeable and verify the current limits before committing a production workflow.

50. MCP / EXTENSION LAYER

The project documents identify these as useful integrations:

Blender MCP
Filesystem MCP
GitHub MCP
Context7 MCP
Supabase MCP
Playwright MCP
Chrome DevTools MCP
Fetch MCP

Use them only when they reduce manual work.

Priority:

1. Filesystem / repo context
2. Context7 for current library docs
3. Blender/cloud asset automation if used
4. GitHub / Supabase for app features
5. Playwright / DevTools for QA

Do not let a connector become a hidden source of visual truth.

The source-of-truth remains the repo manifests and this document.

51. AUTOMATED VISUAL QA

The scene renderer should perform a self-QA pass before presenting a render.

Minimum checks:

1. no console errors
2. not dominated by a single flat color
3. bounding boxes within authored scale
4. camera start position valid
5. camera end position valid

- 6. scene anchors present
- 7. required interactables present
- 8. no missing textures
- 9. no unexpected transparent background assets
- 10. no debug HUD

For every scene compare against:

- approved reference
- previous scene end frame
- next scene start frame

52. CONTINUITY QA SCORE

Give every scene a score out of 100.

Geometry continuity	20
Camera continuity	15
Material continuity	15
Lighting continuity	15
Scale continuity	10
Prop continuity	10
Interaction continuity	10
Atmospheric continuity	5

Do not mark a scene “production ready” below 95.

Any hard failure overrides the score.

Hard failures:

- human visible
- hands visible
- camera not FPP
- wrong major geometry
- wrong color system
- fake text used as actual UI
- teleport detected

interactive object is a flat image
camera clips through geometry

53. THE MOST IMPORTANT CHANGE FROM THE OLD APPROACH

Do not chase “one perfect AI-generated image” forever.

The desired result is:

GENERATED BEAUTY
+
GEOMETRIC TRUTH
+
CAMERA TRUTH
+
INTERACTION TRUTH
+
LIGHTING TRUTH
+
AUDIO TRUTH
+
HAPTIC TRUTH
=
PRESENCE

The image model is one component.

It is not the entire renderer.

54. IMPLEMENTATION ORDER

Phase A — Visual lock

1. lock MASTER_STYLE_BIBLE
2. lock camera language
3. generate SC00
4. accept one canonical image
5. build anchor manifest
6. generate SC01 from SC00
7. continue sequentially through SC06

Do not generate Scenes 07–15 until Scenes 00–06 have continuity approval.

Phase B — Physical substrate

8. create rough scene geometry
9. define anchors
10. create collision volumes
11. define camera path
12. connect transitions

Phase C — Visual realization

13. project generated reference art onto geometry
14. add normal maps
15. add live lights
16. replace live display placeholders
17. add real interactables

Phase D — Interaction

18. door
19. biometric panel
20. table
21. AI core
22. sector stations
23. sticky notes

- 24. globe
- 25. tools / future workstation

Phase E — polish

- 26. audio
- 27. haptics
- 28. focus transitions
- 29. subtle material response
- 30. performance
- 31. accessibility
- 32. semantic fallback

55. DEFINITION OF DONE

A scene is not done because its screenshot is beautiful.

A scene is done when:

the reference image is approved
AND
the geometry exists
AND
the camera path is continuous
AND
the materials match
AND
the lighting is physically motivated
AND
the interactables are real
AND
the transition is causal
AND
the scene survives look-around
AND
the next scene matches its ending state

AND

the weak-device performance is acceptable

AND

the accessibility fallback remains functional

56. FINAL ANTIGRAVITY EXECUTION PROMPT

Implement THE WORKSHOP from the persistent source-of-truth documents.

First read:

1. /docs/workshop-plan-v2.md
2. /docs/scene1-realtime-3d-fix.md
3. /docs/asset-pipeline.md
4. /docs/3d-object-rules.md
5. /docs/THE_WORKSHOP_VISUAL_SCENE_BIBLE_v1.md
6. /docs/notes.md

Then do not code yet.

Report:

- current architecture
- existing scene state
- existing assets
- missing assets
- conflicts between current implementation and this scene bible
- which scene will be built first
- exact continuity anchors involved
- validation steps

Critical rule:

Never substitute a static generated image for a runtime 3D scene.

Use image generation only for approved visual reference and projection-source assets.

Build scenes sequentially.

Start with:

SC00_APPROACH

Do not proceed to SC01 until:

- SC00 reference approved
- SC00 geometry renders
- camera motion works
- look motion works
- door interaction works
- no static background is being used
- screenshot comparison passes

Then continue one scene at a time.

Never silently redesign previous scenes.

Never silently change the Style Bible.

Never add props that are not in the manifest.

Never create fake UI text inside generated imagery.

Every interaction must produce believable physical, visual, auditory, or haptic feedback.

Every transition must have a physical cause.

Every scene must end exactly where the next scene expects it to begin.

The objective is not to make a slideshow that looks cinematic.

The objective is to make the visitor believe the camera occupies a coherent physical place.

57. FINAL CREATIVE DIRECTIVE

The Workshop should not feel like:

a portfolio website with 3D

It should feel like:

a real private engineering environment
that happens to contain a portfolio.

The visitor should progressively infer:

door
machine
facility
intelligence
workshop
work
person

The environment should never explain itself too early.

The strongest visual language is:

physical object first
digital behavior second
text third

The final experience should leave the visitor thinking:

"I was physically there for a minute."

not:

"Someone made a very elaborate WebGL landing page."

That distinction is the entire project.